

Nondistortionary elicitation of beliefs in dynamic problems

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Introduction

- Many experiments involve incentivized truthful reporting of beliefs
 - ▶ examples: self-confidence, cognitive uncertainty, information acquisition, dynamic games,
 - ▶ incentives improve accuracy (Schlag et al. (2015) and many others),
 - ▶ many methods, including, binarized Becker-DeGroot-Marschak (BDM) scoring rule.
- But combined with actions, incentivization may distort behavior:
 - ▶ Example: elicitation of a belief about the grade,
 - ▶ questions about future behavior.
- Distortions are problematic
 - ▶ interpretation of experimental results,
 - ▶ honest instructions (incentivization does not need to be explicit (Danz et al. (2022))),
 - ▶ ethical and legal issues.

Introduction

- **Which questions can be incentivized without distorting behavior? How to ask them?**
- Pęski and Stewart (2025) characterize all incentivizable questions in static problems:
 - ▶ Result: Question must be aligned, i.e., an affine transformation, of payoffs.
 - ▶ Corollary: Question about payoffs is incentivizable.
- **Dynamic problems** have new issues:
 - ▶ richer learning environment (Chambers and Lambert (2021)),
 - ▶ multiple points of asking questions, and
 - ▶ multiple possible points of distortions.
- Different ways actions can affect arrival of new information:
 - ▶ action-independent,
 - ▶ actions affect the quality of information,
 - ▶ actions affect the distribution of states.

Model

Decision problem

- Dynamic decision problem: (A_1, A_2, u)
 - ▶ A_t - finite action set in period t
 - ▶ $u : A_1 \times A_2 \times \Theta \longrightarrow \mathbb{R}$ payoffs
- Expected utility, Bayesian framework $\implies$ expected payoffs:

$$\text{Period 2: } u^*(a_1, p) = \max_{a_2} \mathbb{E}_p u(a_1, a_2, \theta) \text{ for } p \in \Delta\Theta$$

$$\text{Period 1: } U^*(\mu) = \max_{a_1} \mathbb{E}_{\mu(a_1)} u^*(a_1, p) \text{ for } \mu(a_1) \in \Delta\Delta\Theta$$

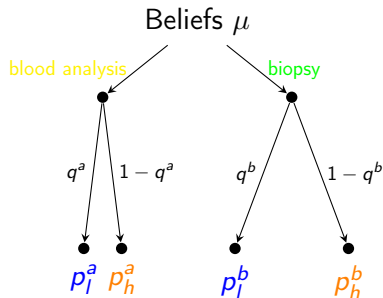
- Actions vs information
 - ▶ Information is not affected by actions: $\mu \in \Delta\Theta$
 - ▶ Actions affect the quality of information: $\mu : A \longrightarrow \Delta\Delta\Theta$ and $\mathbb{E}_{\mu(a)} p = \mathbb{E}_{\mu(b)} p$,
 - ▶ Actions affect states.

Period 1 elicitation

Doctor.

Example: Doctor

- Doctor chooses a method to diagnose tumor count:
 - ▶ blood analysis or biopsy,
 - ▶ biopsy improves information, but it is invasive.
- After observing the outcome, she chooses treatment:
 - ▶ active surveillance or chemotherapy
- Doctor maximizes QuALY.
- Questions
 - ▶ "What are your expected payoffs?"
 - ▶ "How likely will you believe that chemotherapy is optimal?"



$$p_l^b < p_l^a < p_h^a < p_h^b,$$
$$p_0 = q^x p_l^x + (1 - q^x) p_h^x$$

Example: Doctor

Period 1 questions

- $X : A_1 \times \Delta\Theta \longrightarrow \mathbb{R}$ *action-dependent* period 1 question about *period 2 beliefs*.

- ▶ "What are your expected payoffs?" is equivalent to

$$X(a_1, p) = u^*(a_1, p)$$

- ▶ "How likely will you believe that chemotherapy is optimal?"

$$X(a_1, p) = 1\{p > p^*\}.$$

- DM is asked to report $r = \mathbb{E}_{\mu(a_1)} X(a_1, \cdot)$

Example: Doctor

Incentivizability

- Question $X(a, p)$ is *incentivizable in period 1* in dynamic decision problem (A_1, A_2, u) if there exists nondistortionary expansion $(A_1 \times \mathbb{R}, A_2, v)$ such that

no distortions in $t = 1$: $\mathbb{B}_1^v(\mu) \subseteq \mathbb{B}_1^u(\mu) \times \mathbb{R}$,

no distortions in $t = 2$: $\mathbb{B}_2^v(p; a_1, r) \subseteq \mathbb{B}_2^u(a_1, p)$

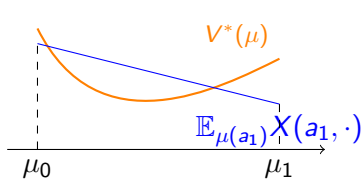
truthful reporting: $\mathbb{B}_1^v(\mu) \subseteq \{(a_1, r) : r = \mathbb{E}_{\mu(a_1)} X(a_1, p)\}$

- ▶ $\mathbb{B}_1(\mu)$ - the set of best responses in period 1
 - ▶ $\mathbb{B}_2(p; a_1, r)$ - the set of best responses in period 2 following action a_1
 - ▶ $r \in \mathbb{R}$ - period 1 report of the expectation of a variable $X(a, p)$
 - ▶ in the paper, we also allow for additional finitely many supporting actions (example: Chambers and Lambert (2021))
- In principle, $v(a_1, a_2, \theta)$ is independent from u , but typically it is related.
- Assumption: one question only!

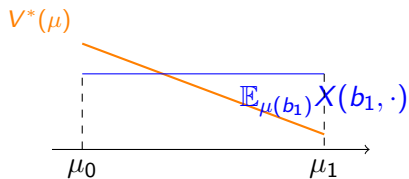
Example: Doctor

Value matching

- Value function $V^*(\mu) = \max_{a_1, r} \mathbb{E}_\mu v^*(a_1, r, \cdot)$ is convex.



changing beliefs $\implies$ str. convex value

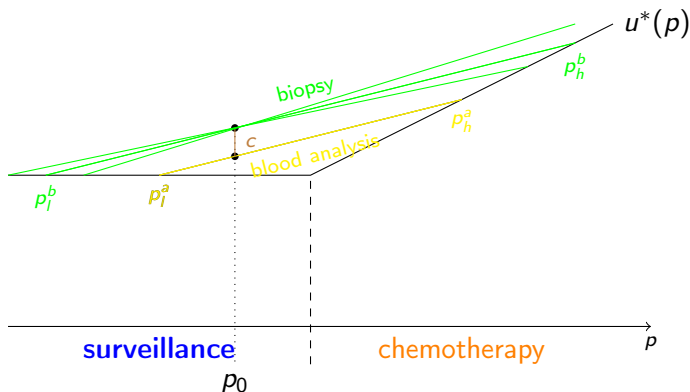


constant beliefs $\implies$ affine value

- If a_1 and b_1 are best responses at beliefs μ_0 and μ_1 , then such configuration of beliefs imply that X cannot be incentivizable.
- We can generate such a pair of beliefs for the question about chemotherapy, but not for the question about the payoffs.

Example: Doctor

Value matching



- There are beliefs μ_0 and μ_1 that keep the indifference between two diagnostics choices, and keep the probability of chemotherapy after blood analysis constant, but not after biopsy.

Period 2 elicitation

Student.

Example: Student

- Student decides whether to write an *optional* test with a single True-False question.
- If he stays out, his grade is $h = 80$.
If he writes the test, its weight is 25%.
- The decision is made before the Student sees the test, which reveals information about the difficulty.
- Question "What is your expected grade/score?"

$$X(a_1, a_2, \theta) = u(a_1, a_2, \theta), \text{ or}$$
$$X(a_1, a_2, \theta) = P(a_2, \theta).$$

- $\Theta = \{0, 1\}$
- $A_1 = \{0, 1\}, A_2 = \{0, 1\}$
- test score:

$$P(a_2, \theta) = 1\{a_2 = \theta\}$$

- payoffs:

$$u(a_1, a_2, \theta) = h + \frac{1}{4}a_1(P(a_2, \theta) - h)$$

Example: Student

Incentivizability in period 2

- Question $X(a, p)$ is *incentivizable in period 2* in dynamic decision problem (A_1, A_2, u) if there exists nondistortionary expansion $(A_1, A_2 \times \mathbb{R}, v)$ such that

no distortions in $t = 1$: $\mathbb{B}_1^v(\mu) = \mathbb{B}_1^u(\mu)$,

no distortions in $t = 2$: $\mathbb{B}_2^v(a_1, p) \subseteq \mathbb{B}_2^u(a_1, p) \times \mathbb{R}$

truthful reporting: $\mathbb{B}_1^v(\mu) \subseteq \{(a_1, r) : r = \mathbb{E}_p X(a_1, a_2, \theta)\}$

- $r \in \mathbb{R}$ - period 1 report of the expectation of a variable $X(a_1, a_2, \theta)$
- Incentivization in period 2 increases value of information, which may distort behavior in period 1.

Example: Student

Value of information

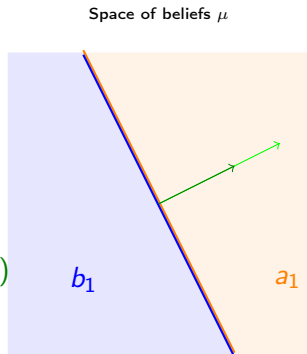
- If X is *incentivizable*, the first-period indifference condition must be preserved:

$$\mathbb{B}^u(\mu) = \{a_1, b_1\} = \mathbb{B}^v(\mu).$$

- Hence, vectors

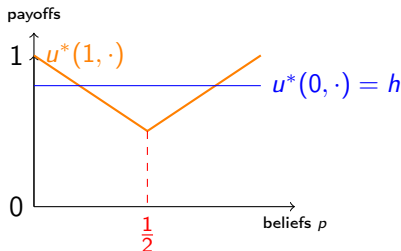
$$u^*(a_1, \cdot) - u^*(b_1, \cdot) \sim v^*(a_1, \cdot) - v^*(b_1, \cdot)$$

are collinear.



Example: Student

Period 2 necessary conditions



- $u^*(1, p) - u^*(0, p)$ is strictly convex **only at** $p = \frac{1}{2}$.
- $\mathbb{E}_p X(1) = u^*(1, p)$ is changing $\implies v^*(1, p)$ is str. convex at each p .
- $\mathbb{E}_p X(0) = u^*(0, p)$ is changing $\implies v^*(0, p)$ is affine at each p .
- $v^*(1, p) - v^*(0, p)$ is str. convex **at each** p .

Complete characterization

Incentivizable questions $\times$:

	Action-independent information	Actions may affect the quality of information	Actions may affect the states
$t = 1$	$\gamma u^*(a_1, p) + f(p)$ PS25 conditions	$\gamma u^*(a_1, p) + l(p)$ nontrivial continuation	$\gamma u^*(a_1, p)$ always
$t = 2$	$\gamma u_2(a_2, \theta) + d(\theta)$ only if payoffs are <u>separable</u>	$\times$	$\times$

- $\gamma = 0, 1$
- $f(p)$ is arbitrary (sufficiently regular) function of beliefs
- $l(p)$ is affine in p